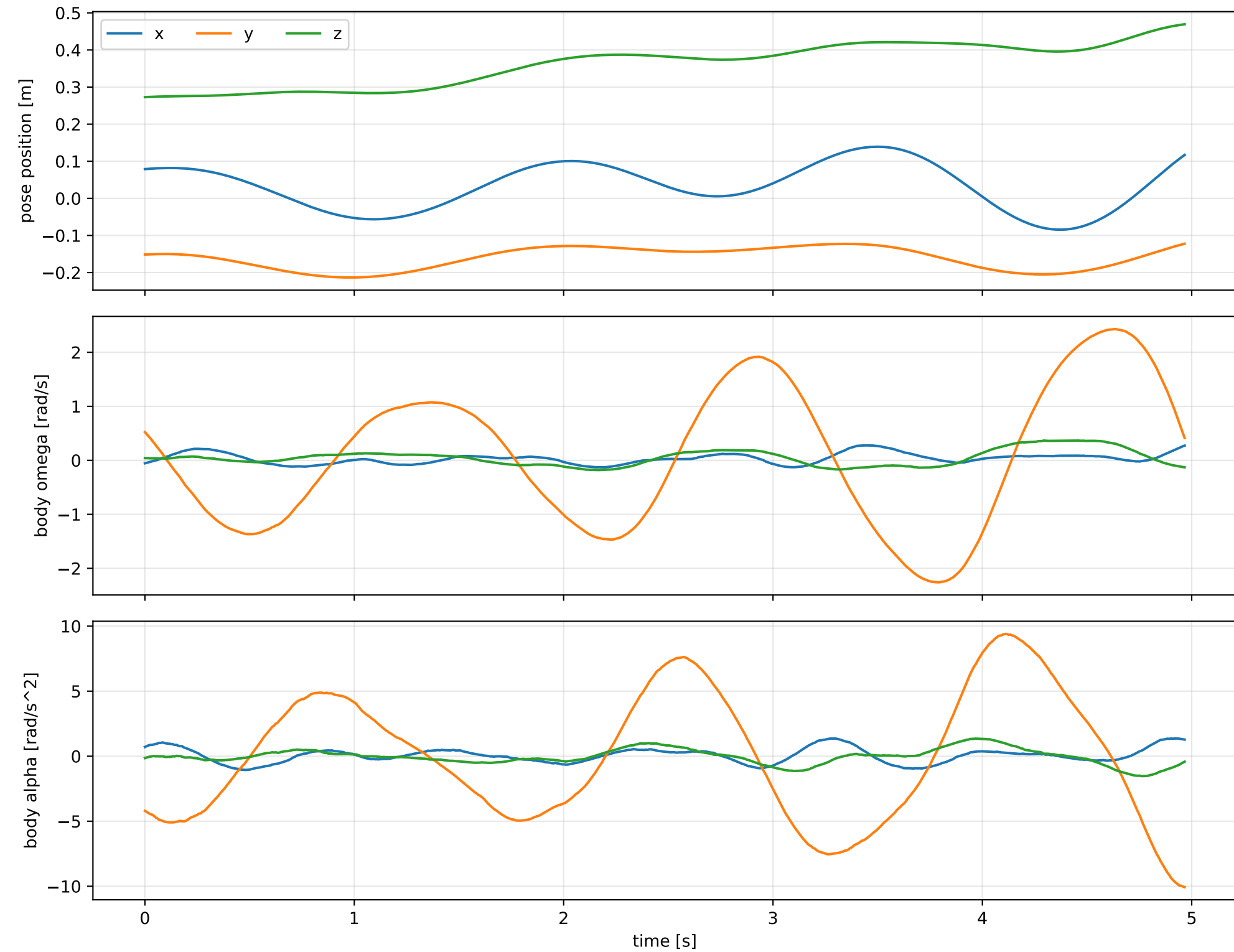
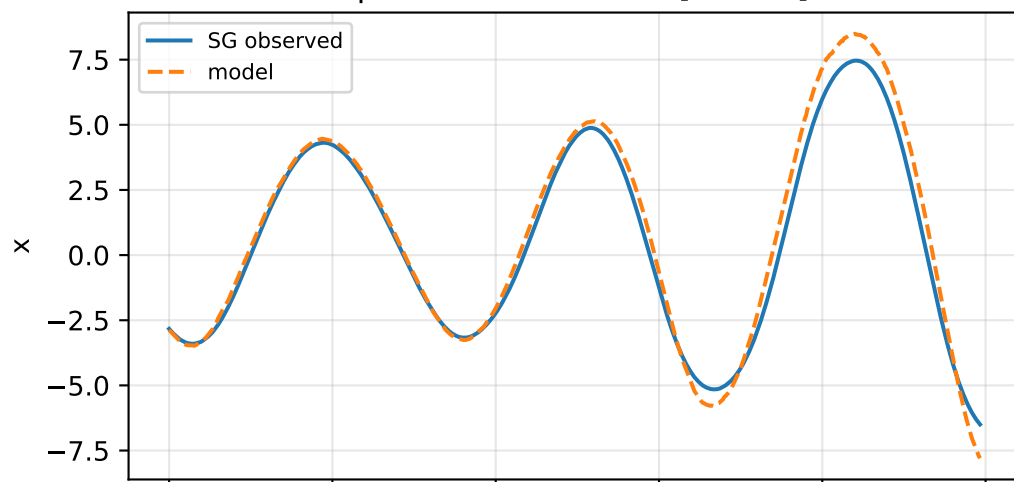


lag_zero: SG trajectory and derived motion

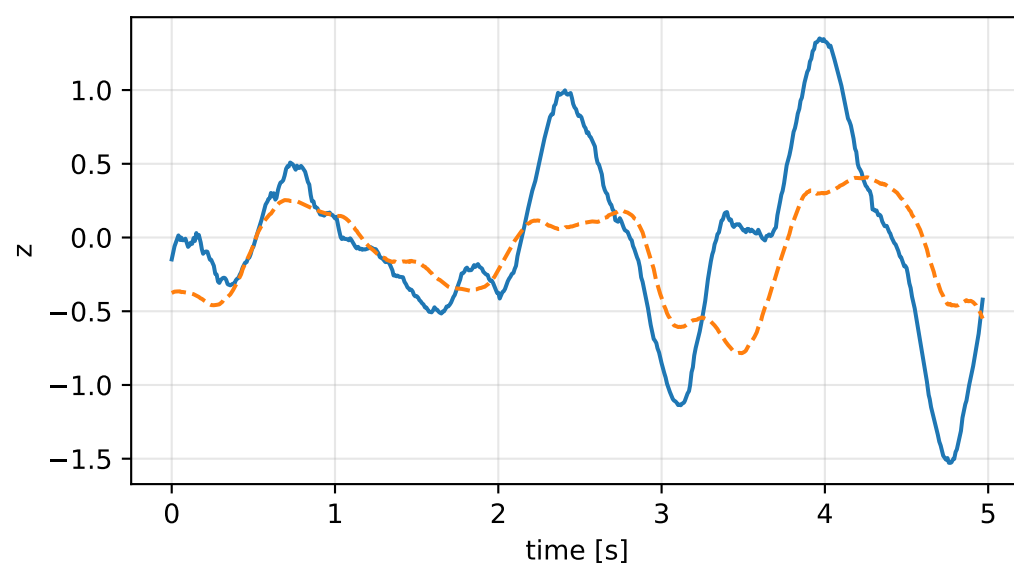
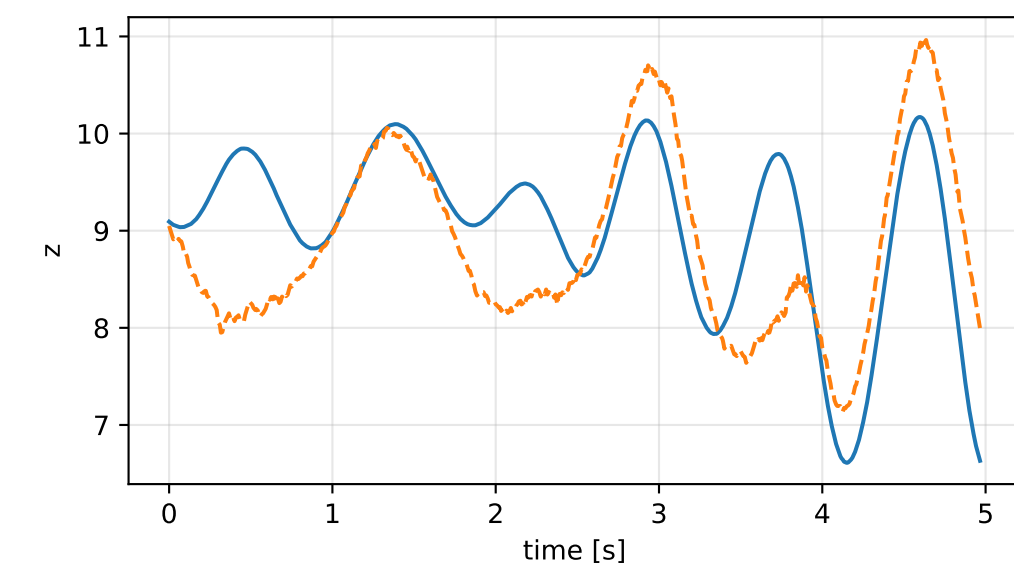
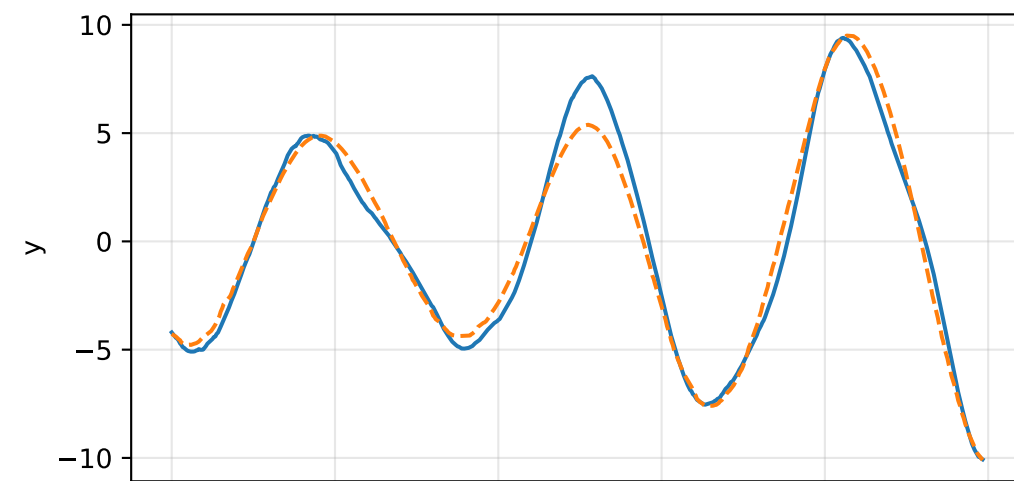
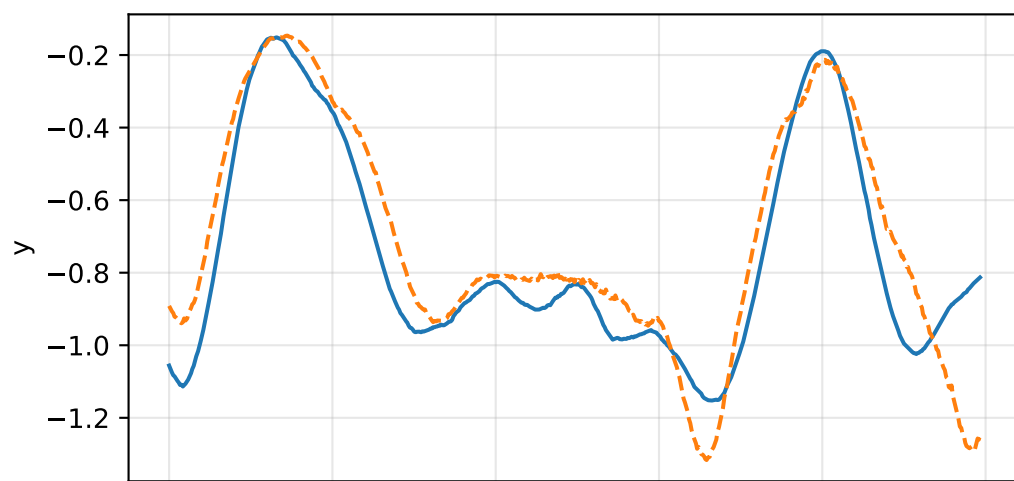
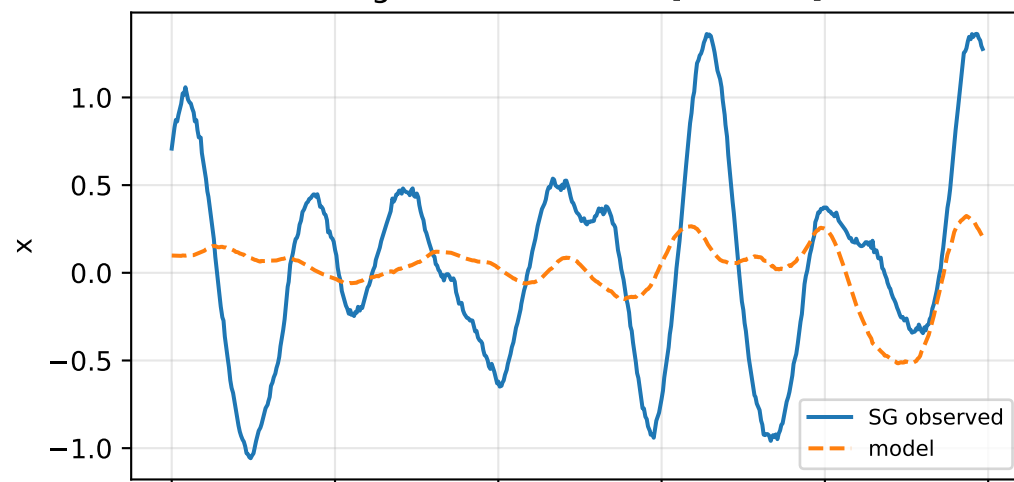


lag_zero: acceleration residual objective

specific acceleration [m/s^2]

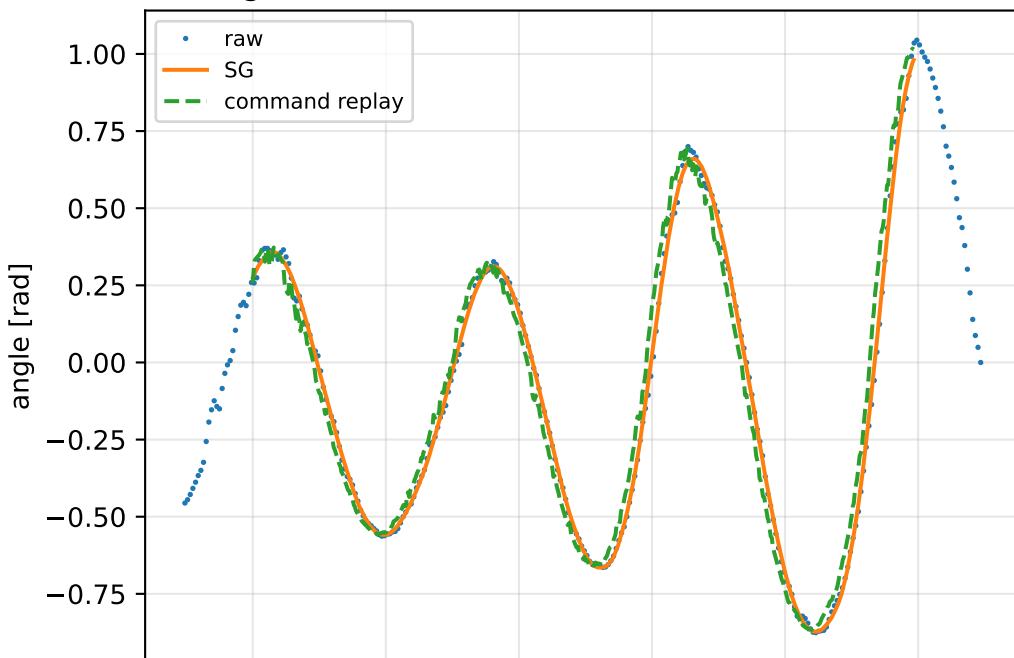


angular acceleration [rad/s^2]

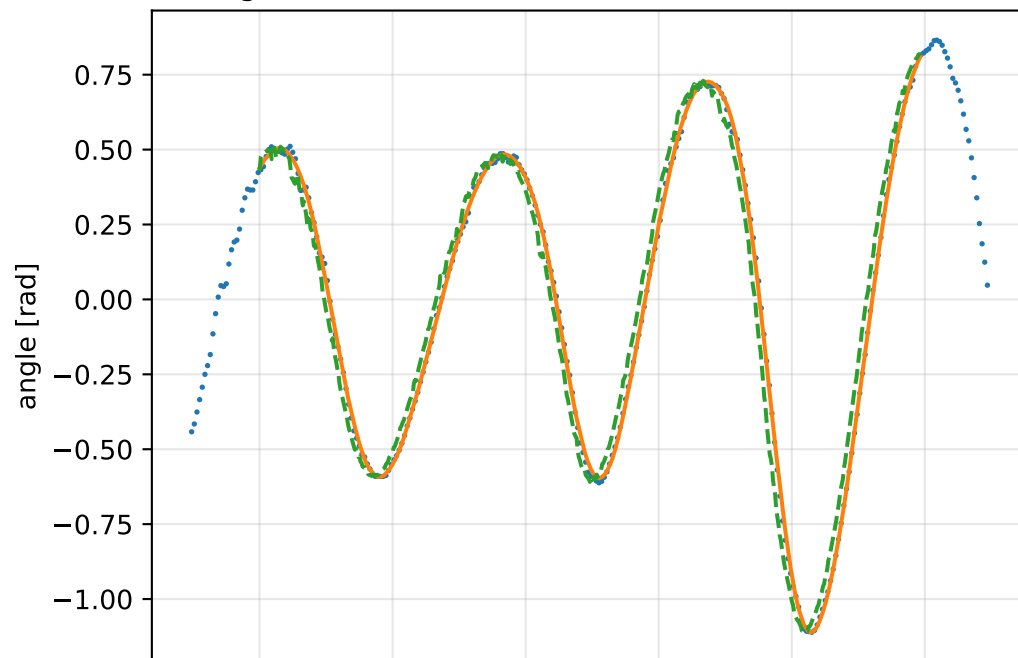


lag_zero: gimbal measurement audit (objective=measured_sg)

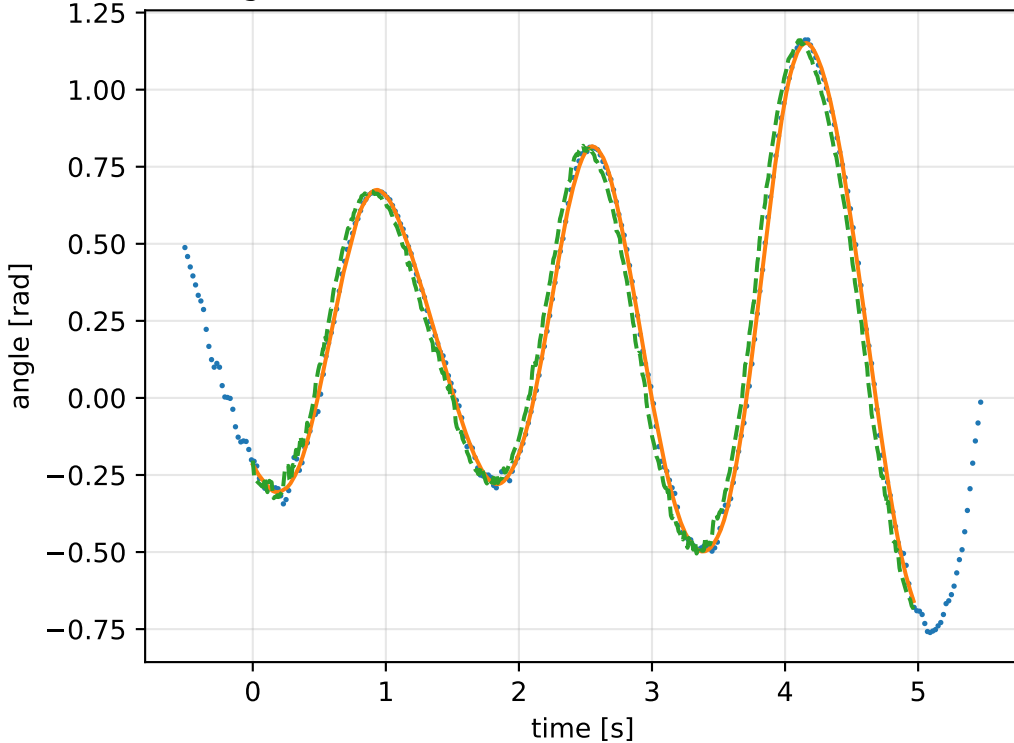
gimbal 1: RMSE=0.0759, max=0.2095 rad



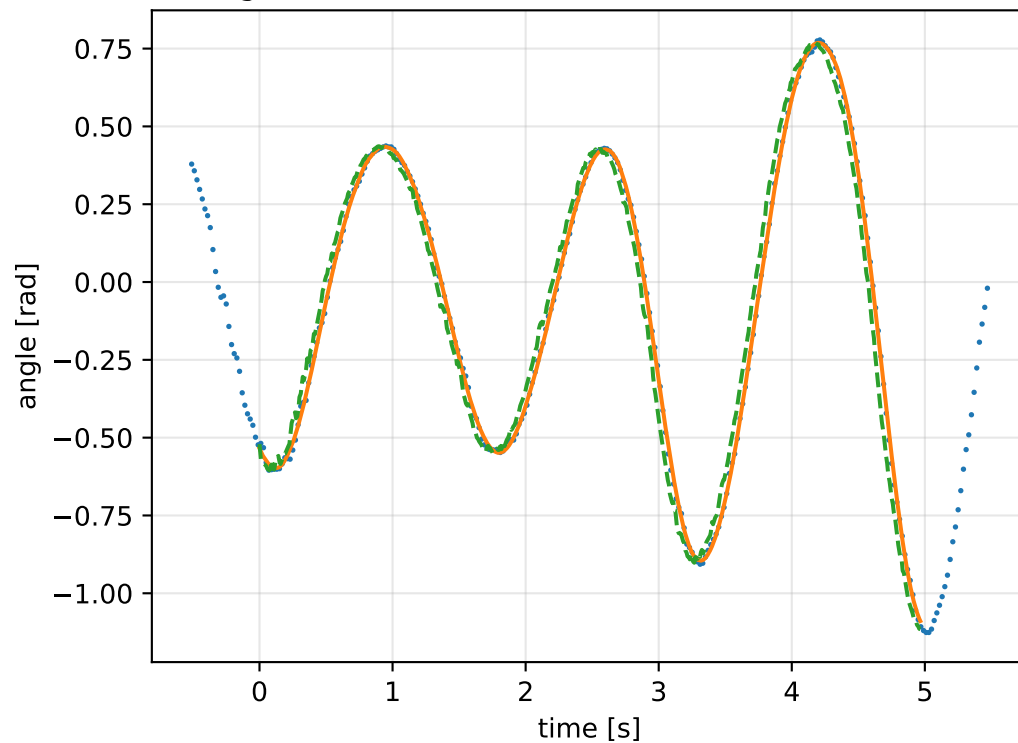
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

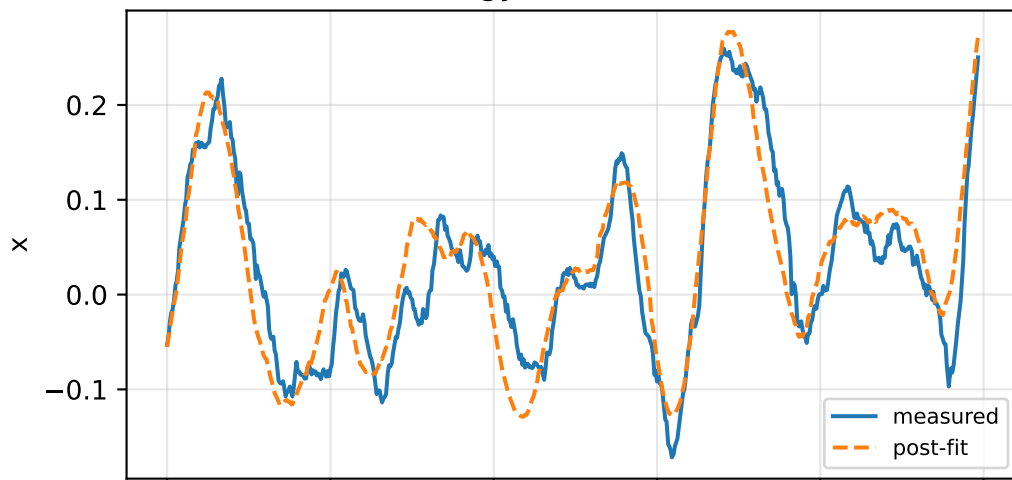


gimbal 4: RMSE=0.07131, max=0.1737 rad

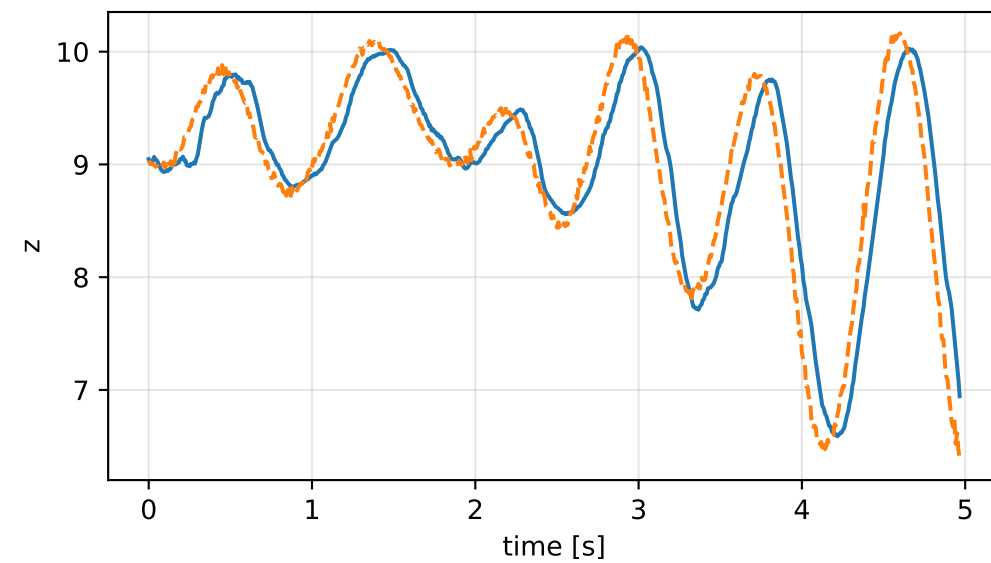
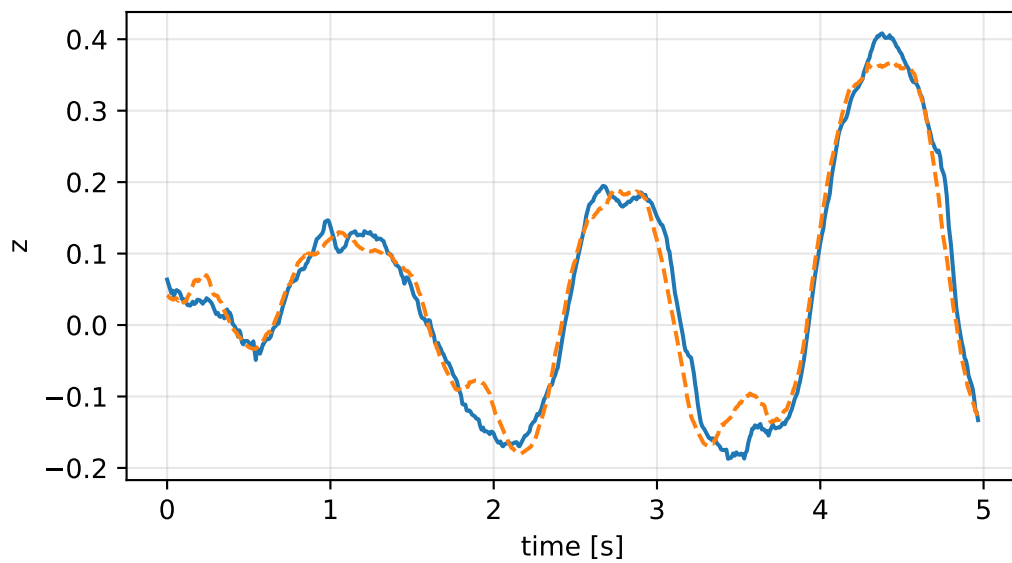
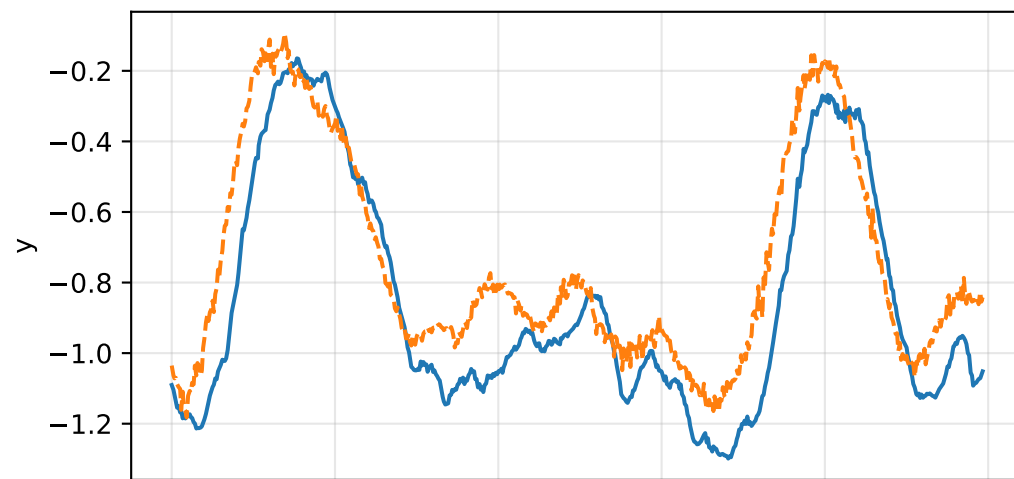
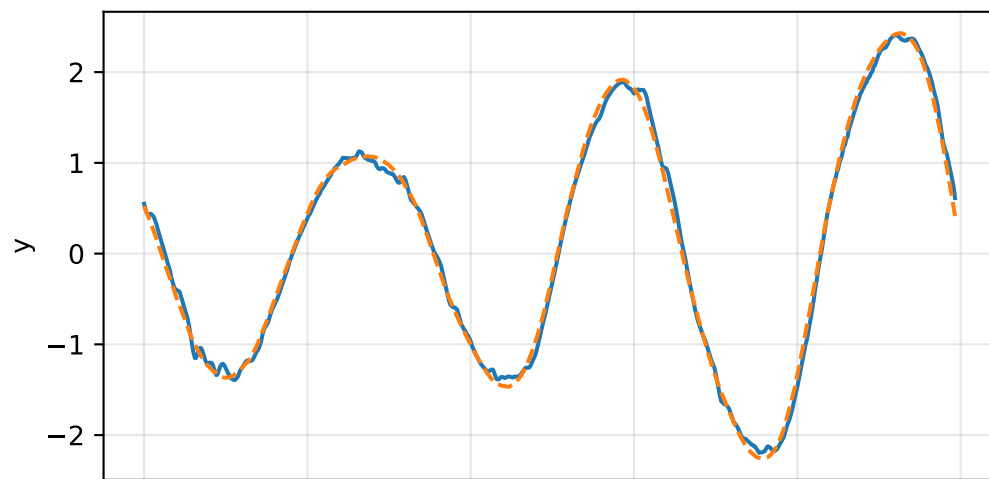
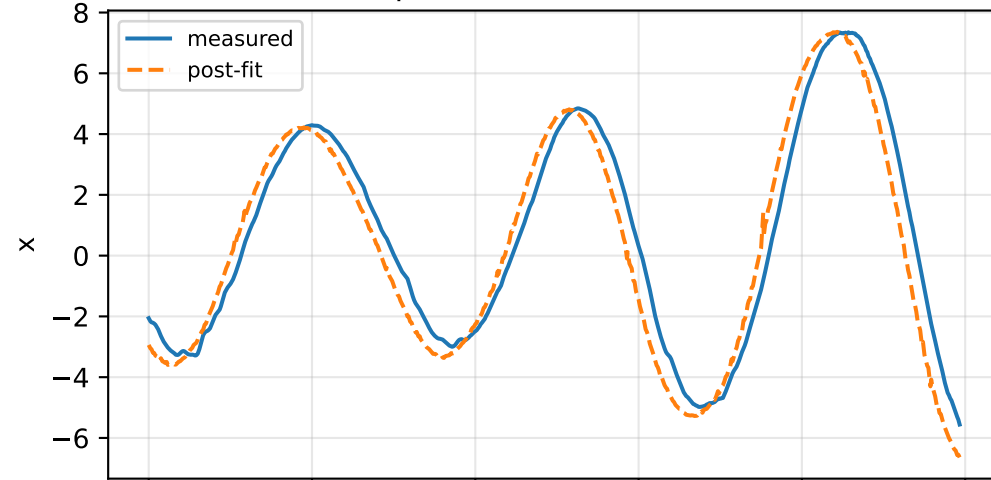


lag_zero: IMU comparison (diagnostic only)

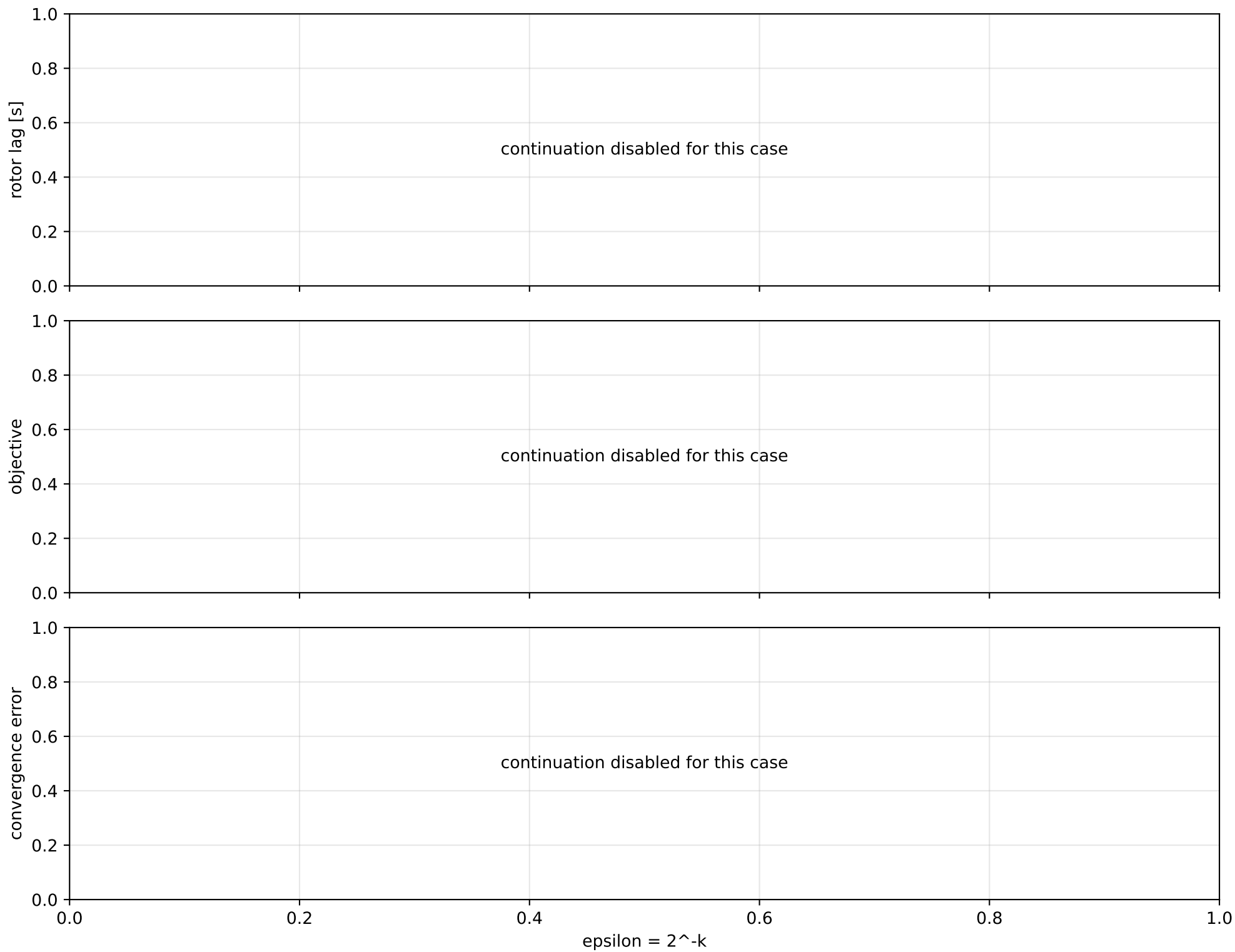
gyro [rad/s]



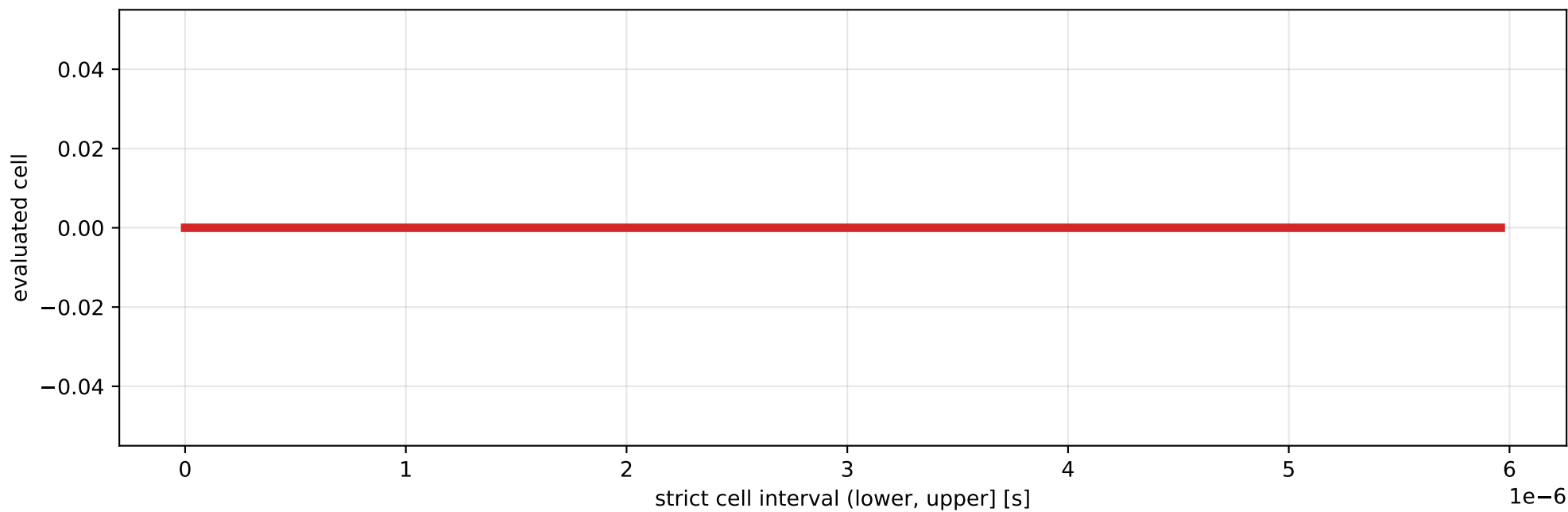
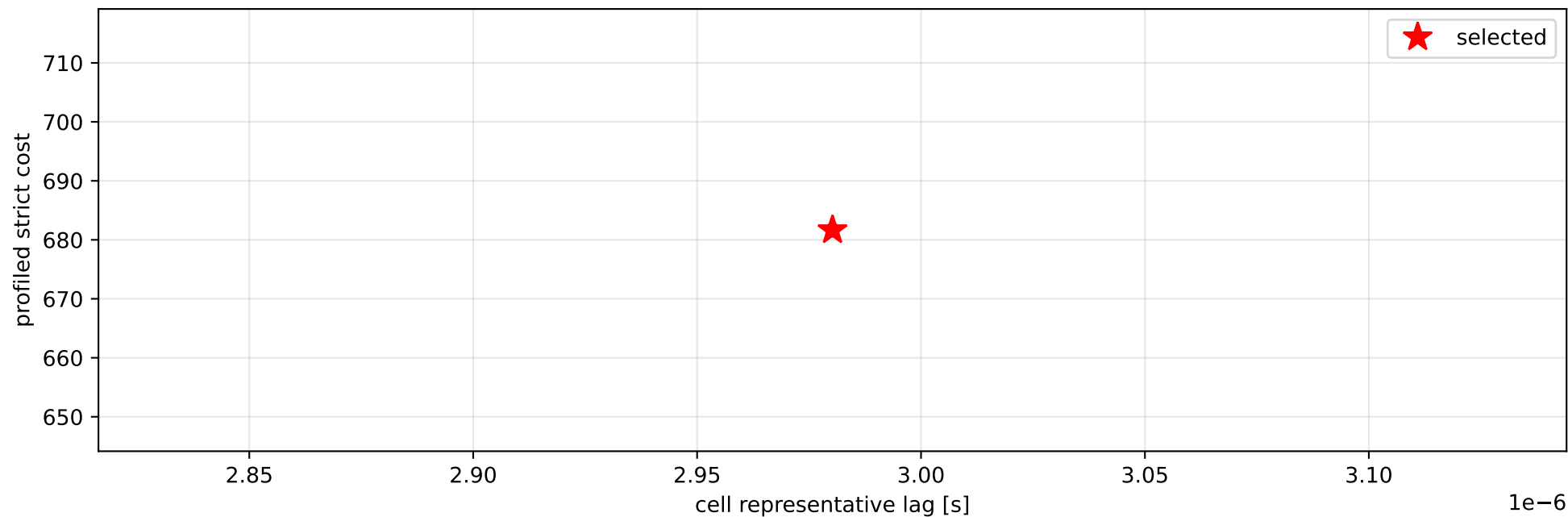
specific force [m/s²]



lag_zero: smooth-to-strict continuation



lag_zero: exact strict-ZOH lag cells



selected (0, 5.96046448e-06] s; final smooth 0 s; data boundary=False

lag_zero: parameter estimate

Case: lag_zero

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 6.64417504

inertia [kg m²]:

```
[[ 9.56284766  0.07165215  0.07179355]
 [ 0.07165215  0.19355214 -0.5049335 ]
 [ 0.07179355 -0.5049335   9.44208089]]
```

CoG body [m]: [0.21493105 -0.00516985 -0.0475312]

force effectiveness: [4.67018608e-04 4.97548540e+00 4.55468149e+00 1.57213476e-01]

scale-free J/m [m²]:

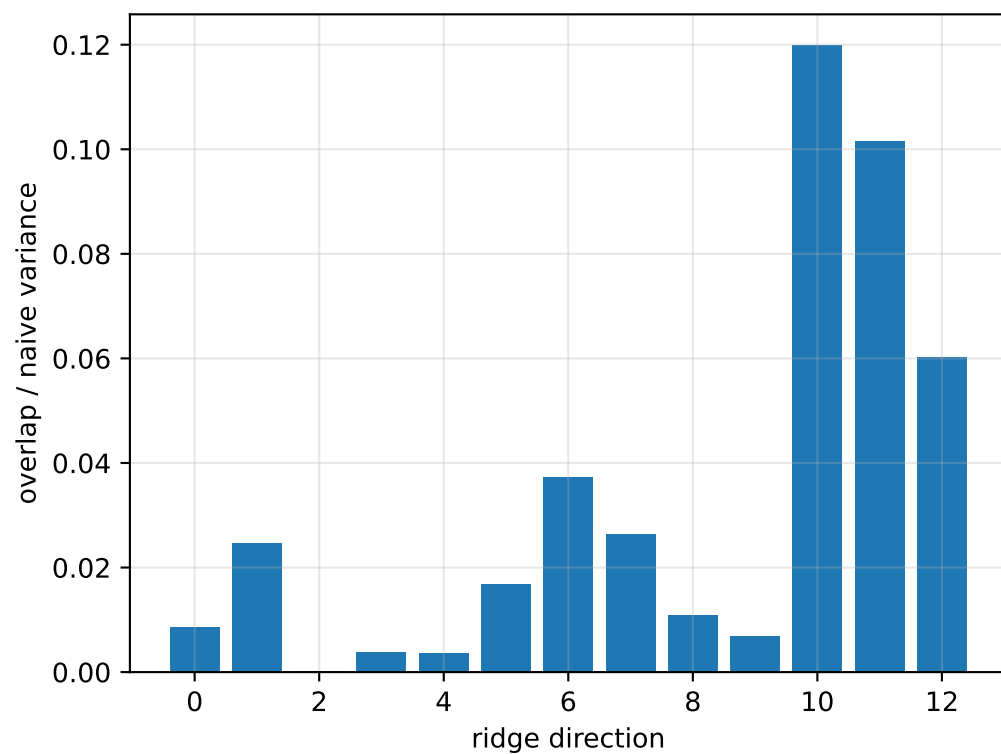
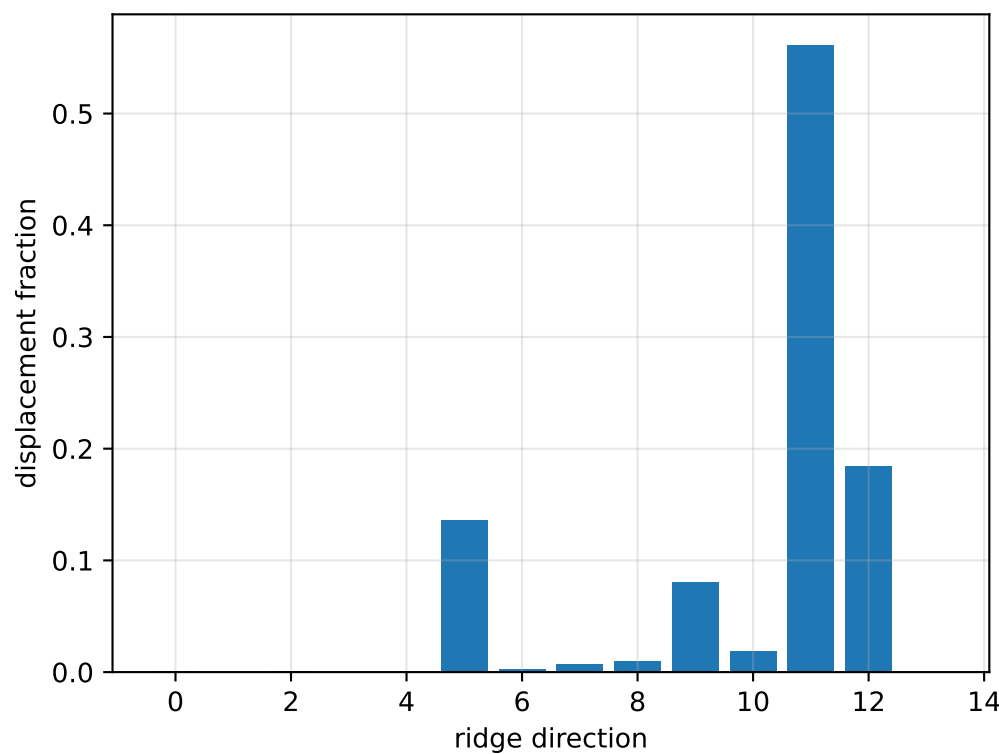
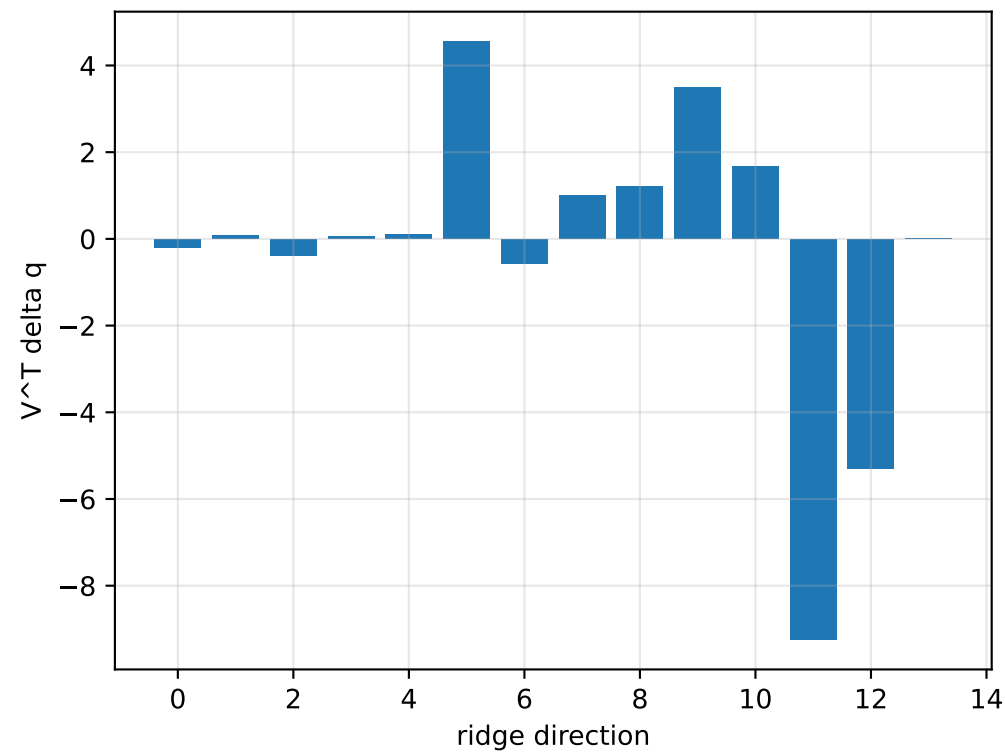
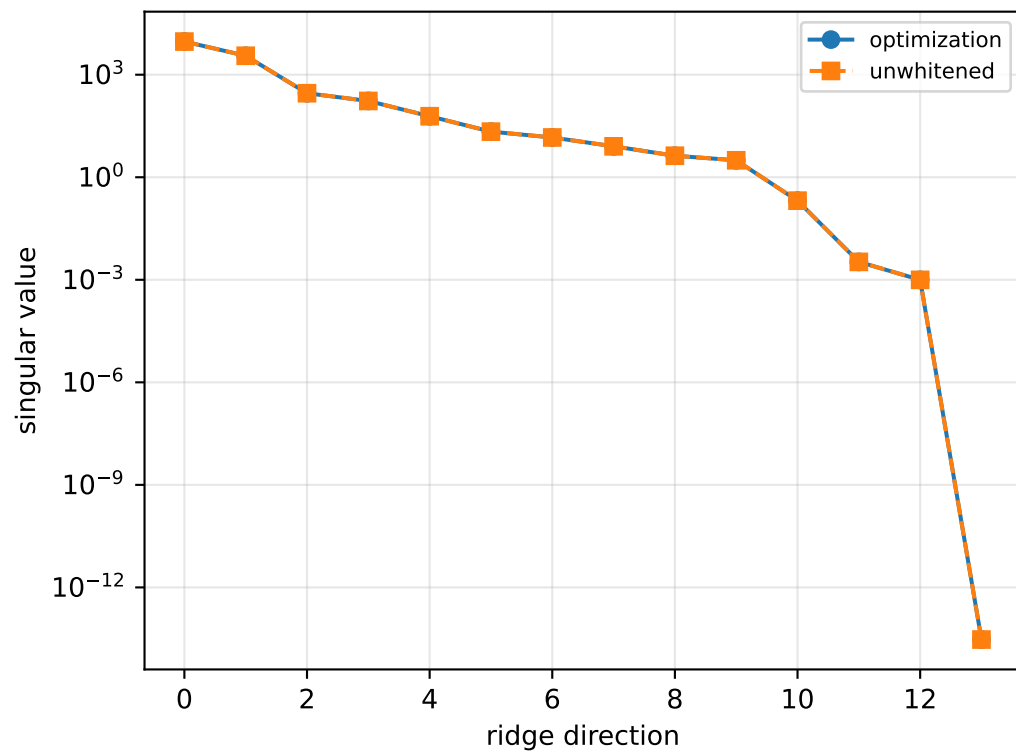
```
[[ 1.43928292  0.01078421  0.01080549]
 [ 0.01078421  0.0291311  -0.07599642]
 [ 0.01080549 -0.07599642  1.42110658]]
```

scale-free f/m: [7.02899314e-05 7.48849235e-01 6.85514975e-01 2.36618504e-02]

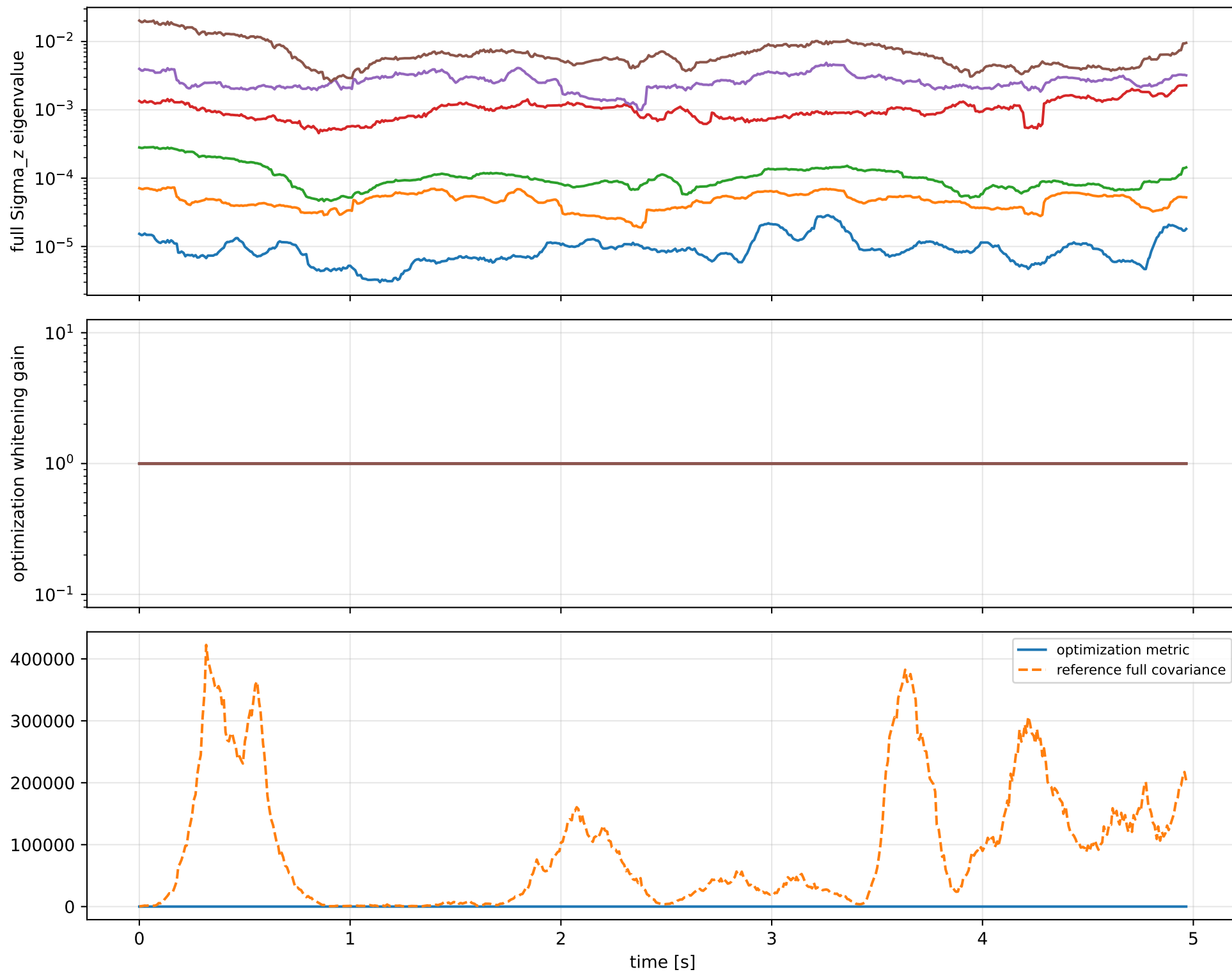
rotor lag cell: (0, 5.96046448e-06] s

representative: 0 s

lag_zero: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=8.26\text{e-}13$)



lag_zero: optimization vs reference covariance



lag_zero: wrench / closure (force max 2.31e-14, torque max 3.55e-15)

